

PROJECT BUTTON HR–V0 DXL PROTECTION CARRIER

Single–channel R155 measurement carrier; assemble three controlled variants; no robot–baseline change.

01 Single–channel DXL branch protection core

File: 01_protection_core.kicad_sch

02 UVLO and OVLO threshold candidates

File: 02_threshold_dividers.kicad_sch

03 Bypass, slew and transient candidates

File: 03_bypass_and_transients.kicad_sch

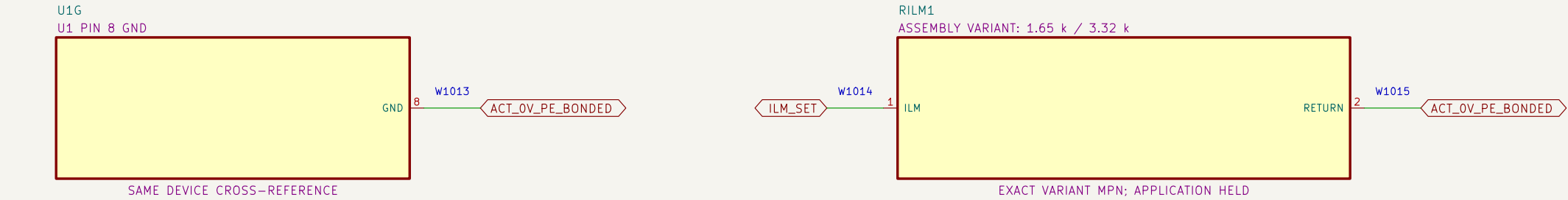
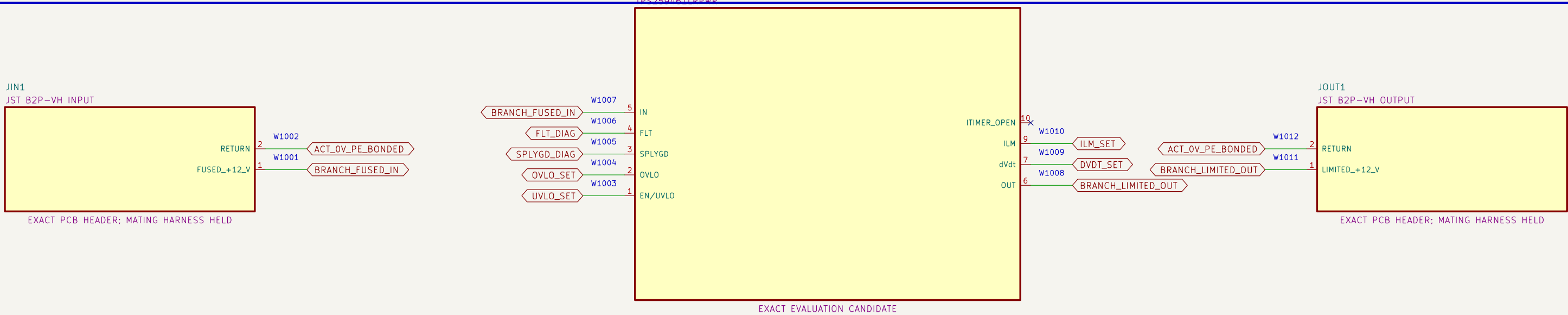
04 Measurement and diagnostic points

File: 04_measurement_points.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /		
File: hr–v0–dxl–protection–carrier–p0.2.kicad_sch		
Title: PROJECT BUTTON HR–V0 DXL PROTECTION CARRIER index		
Size: A3	Date: 2026–08–09	Rev: DXL–PROT–CARRIER–P0.2
KiCad E.D.A. 10.0.5	Id: 1/5	

01 Single-channel DXL branch protection core

Exact active, power boundaries and assembly-variant current setting.



NOTE 1: TPS259461L limits forward current only; reverse current is unbounded while ON.

NOTE 2: ITIMER is deliberately open. SPLYGD and FLT are unpulled diagnostic test points only.

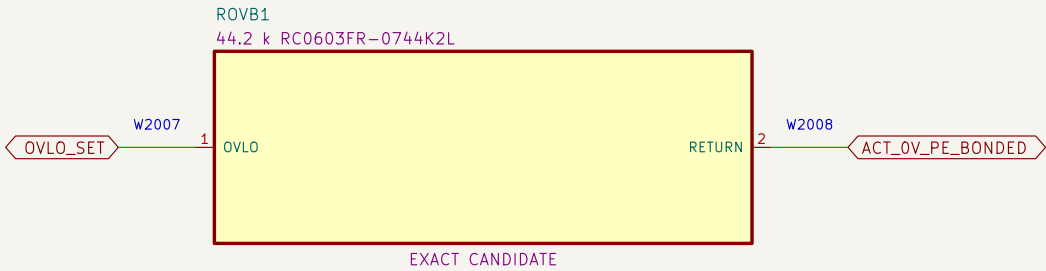
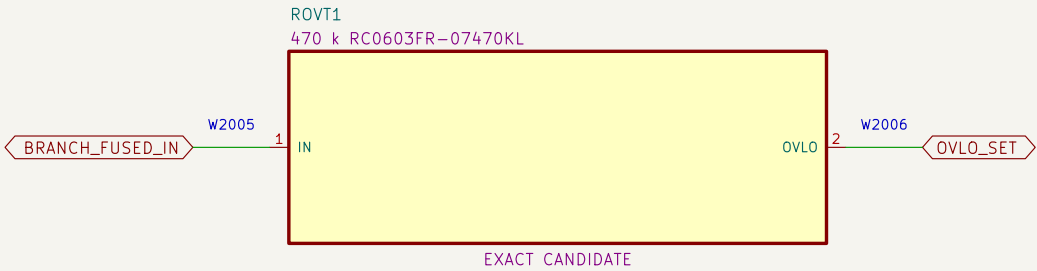
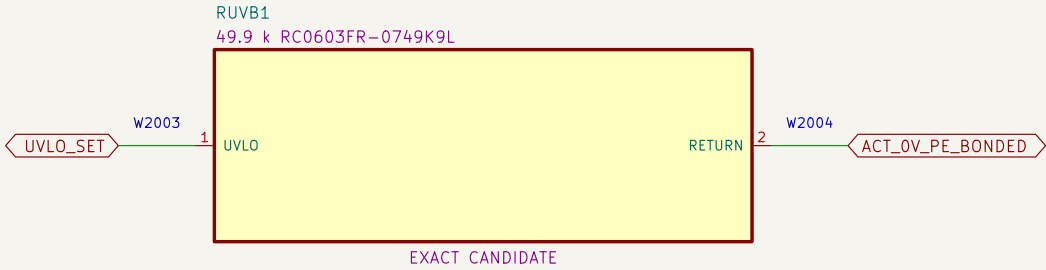
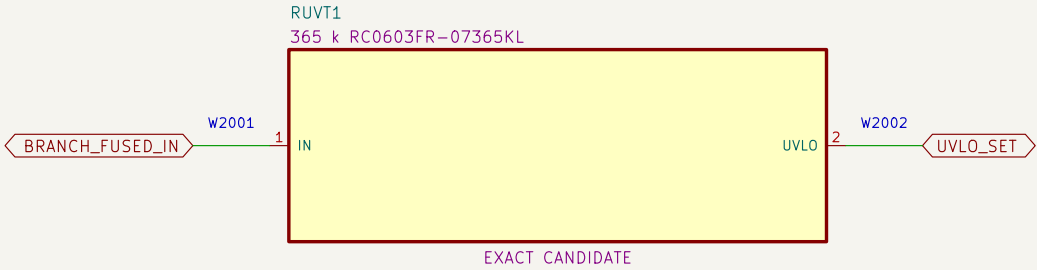
NOTE 3: J1/J2 assemble RILM=1.65 kOhm; G1 assembles RILM=3.32 kOhm. Never substitute variants without a controlled record.

NOTE 4: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /01 Single-channel DXL branch protection core/ File: 01_protection_core.kicad_sch		
Title: PB HR-VO DXL-PROT-CARRIER-P0.2 – 01		
Size: A3	Date: 2026-08-09	Rev: DXL-PROT-CARRIER-P0.2
KiCad E.D.A. 10.0.5		Id: 2/5

02 UVLO and OVLO threshold candidates

Exact resistor identities; divider tolerance and physical thresholds remain open.



NOTE 1: RUVT1/RUVB1 create the nominal UVLO candidate; ROVT1/ROVB1 use TI's 14 V application—example values.

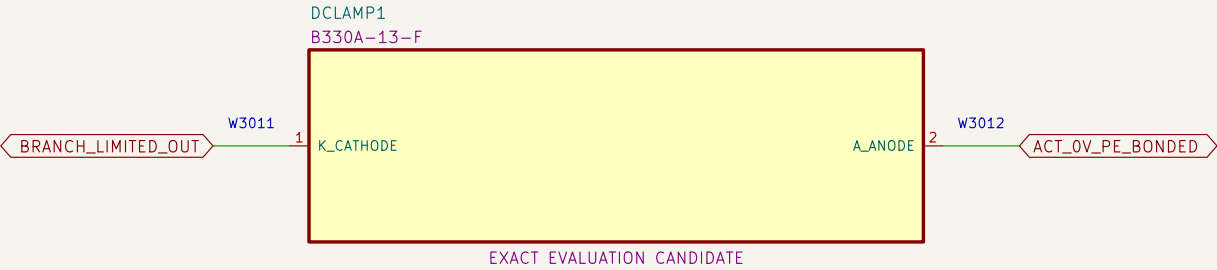
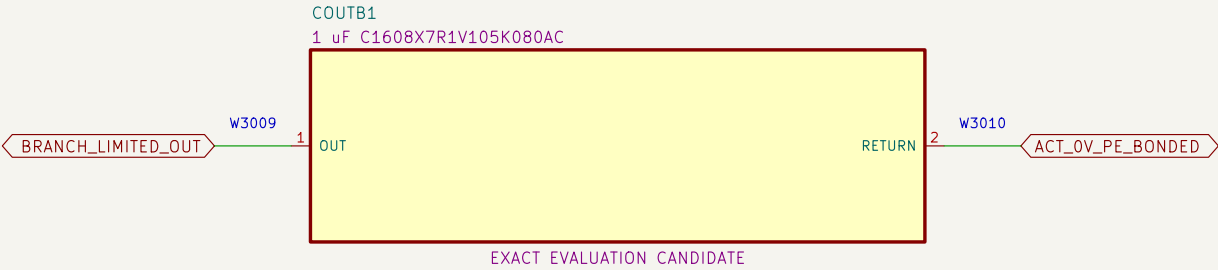
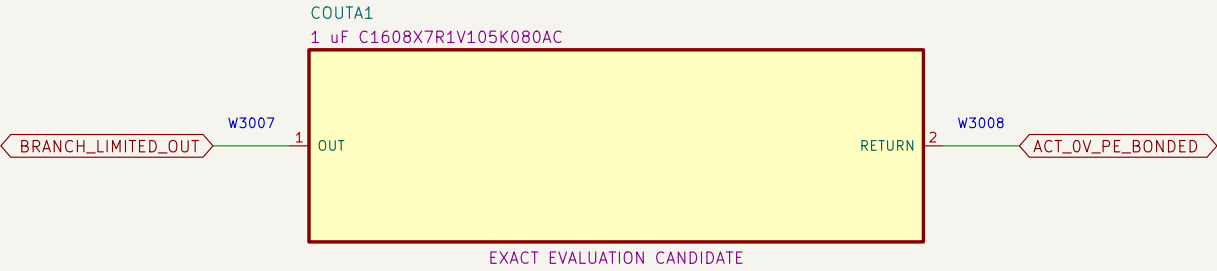
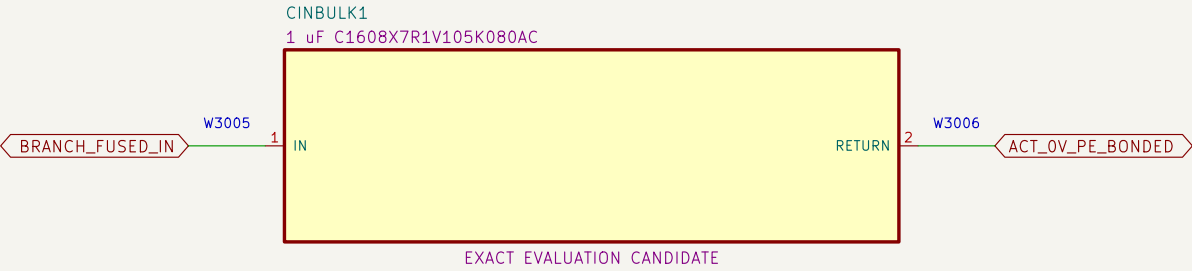
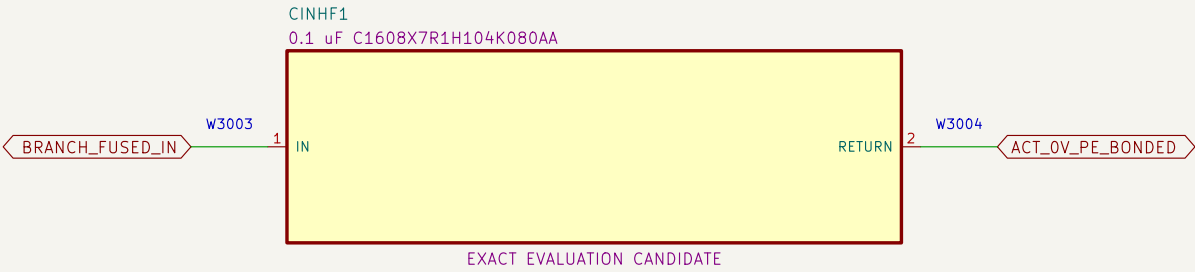
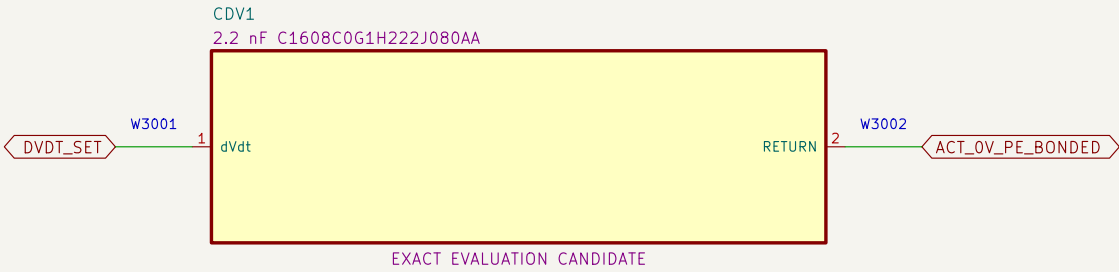
NOTE 2: Received resistor value, tolerance, temperature drift and measured thresholds remain test gates.

NOTE 3: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /02 UVLO and OVLO threshold candidates/ File: 02_threshold_dividers.kicad_sch		
Title: PB HR-V0 DXL-PROT-CARRIER-P0.2 - 02		
Size: A3	Date: 2026-08-09	Rev: DXL-PROT-CARRIER-P0.2
KiCad E.D.A. 10.0.5	Id: 3/5	

03 Bypass, slew and transient candidates

Exact passive identities; physical transient, effective–capacitance and thermal proof remain open.



NOTE 1: CINHF1 and CINBULK1 implement TI input guidance; COUTA1/COUTB1 provide nominal output–capacitance margin only.

NOTE 2: DCLAMP1 is the local output Schottky. The external regenerative shunt candidate is not installed on this carrier.

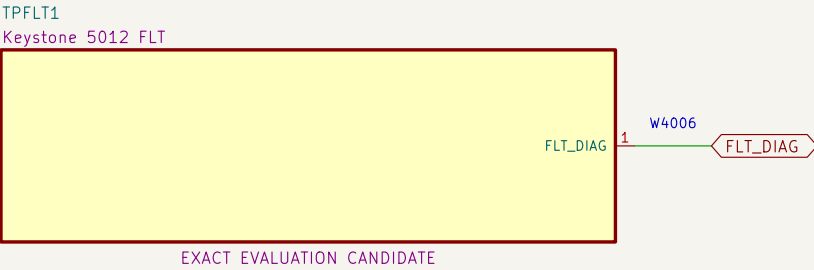
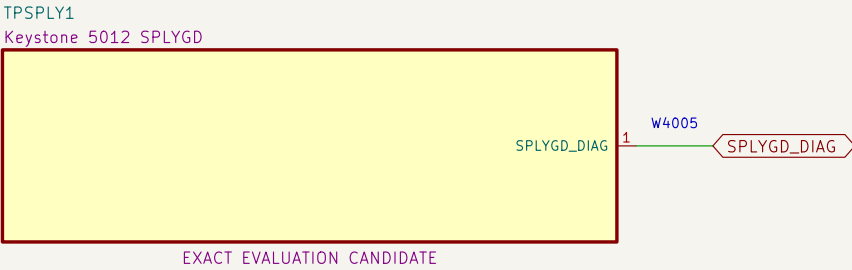
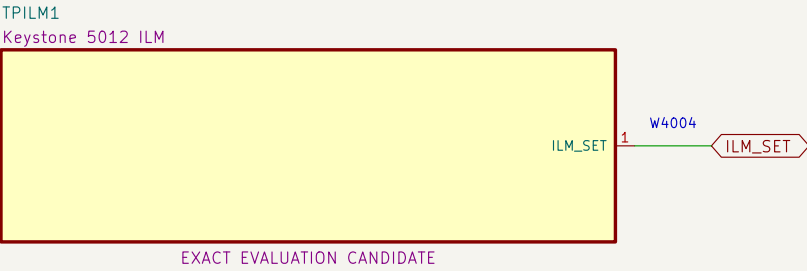
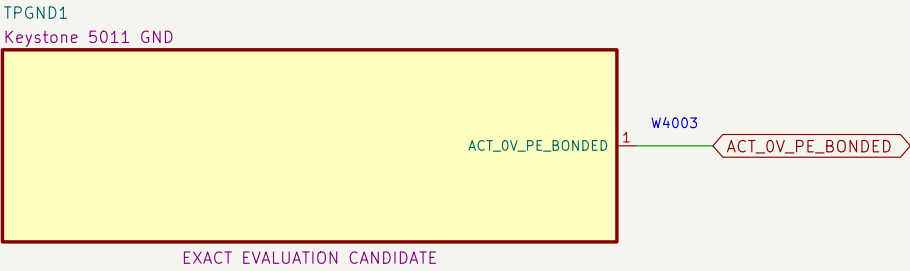
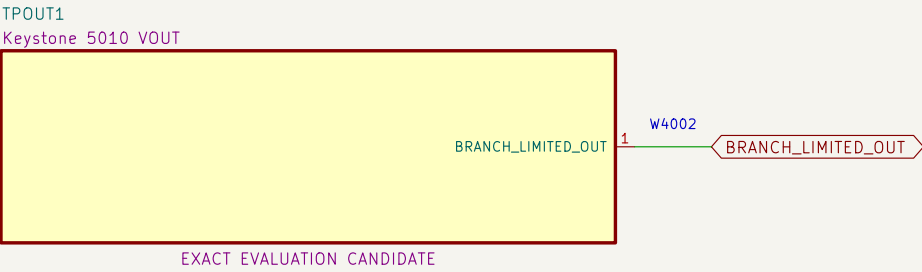
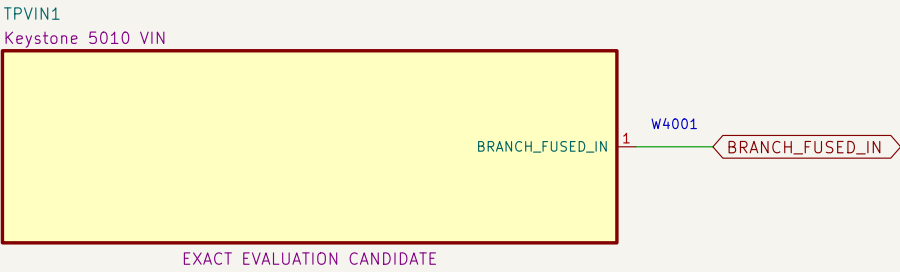
NOTE 3: TVS need and identity remain SELECTION REQUIRED pending actual interconnect–

NOTE 4: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /03 Bypass, slew and transient candidates/ File: 03_bypass_and_transients.kicad_sch		
Title: PB HR–VO DXL–PROT–CARRIER–P0.2 – 03		
Size: A3	Date: 2026–08–09	Rev: DXL–PROT–CARRIER–P0.2
KiCad E.D.A. 10.0.5	Id: 4/5	

04 Measurement and diagnostic points

All measurements require a released method; no diagnostic has safety or motion authority.



NOTE 1: ILM total parasitic and probe capacitance must remain below TI's 50 pF limit.

NOTE 2: SPLYGD and FLT are unpulled open-drain observations only; no safety, reset or motion credit is assigned.

NOTE 3: Test-plan and data records are blank. No physical test has been executed.

NOTE 4: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /04 Measurement and diagnostic points/
File: 04_measurement_points.kicad_sch

Title: PB HR-V0 DXL-PROT-CARRIER-P0.2 – 04

Size: A3	Date: 2026-08-09	Rev: DXL-PROT-CARRIER-P0.2
KiCad E.D.A. 10.0.5	Id: 5/5	